MA 5755: Data Analysis & Visualization in R/Python/SQL

Week 9: Bayesian and Nonparametric Regression with Gaussian Processes

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Spring 2026

Outline

1. Nonparametric Regression
2. Bayesian Linear Regression
3. Kernel Perspective
4. Gaussian Processes
5. Connections to Modern Machine Learning

Regression Beyond Parametric Models

We consider:

$$y = f(x) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Parametric approach:

- Assume $f(x) = w^T x$ or polynomial form
- Fixed finite-dimensional parameterization

Limitation:

- Model misspecification if true f is complex
- Global structure imposed a priori

Nonparametric viewpoint:

- Do not assume functional form
- Estimate $f(x)$ locally from data
- Complexity grows with sample size

Key question: How to estimate $f(x)$ using nearby observations?

Local Averaging and Kernel Smoothers

General estimator:

$$\hat{f}(x) = \sum_{i=1}^{n} w_i(x) y_i, \quad w_i(x) \geq 0, \ \sum_i w_i(x) = 1$$

Nadaraya–Watson estimator:

$$\hat{f}(x) = \frac{\sum_{i=1}^{n} K(x, x_i)\, y_i}{\sum_{i=1}^{n} K(x, x_i)}$$

Interpretation:

- $K(x, x_i)$ measures similarity between x and x_i
- Prediction is a locally weighted average
- Nearby points dominate due to higher weights

Typical kernel:

$$K(x, x_i) = \exp\left(-\frac{\|x - x_i\|^2}{2h^2}\right)$$

Observation: This resembles a data-dependent basis expansion.

Bandwidth and Bias–Variance Structure

Bandwidth h determines the effective neighborhood.

Effects:

- Small h: highly local, captures fine structure, high variance
- Large h: smoother estimate, high bias

Bias–variance tradeoff:

$$\text{MSE} = \text{Bias}^2 + \text{Variance}$$

Analogy:

- h plays role similar to regularization parameter λ
- Both control effective model complexity

Limitations of kernel smoothing:

- No uncertainty quantification
- Bandwidth selection is heuristic
- Not derived from a probabilistic model

From Kernel Smoothing to Structured Models

Kernel smoothing suggests:

- Predictions depend on similarities $K(x, x_i)$
- No explicit parameter vector is learned

Limitation:

- Lacks global coherence and uncertainty estimates

Motivating question:

Can we build a model where:

- similarity is encoded via kernels, and
- predictions come with uncertainty?

Answer: Move to a Bayesian framework.

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Bayesian Linear Regression

Model:

$$y = Xw + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 I)$$

Prior:

$$w \sim \mathcal{N}(0, \Sigma_p)$$

Posterior:

$$p(w|X, y) \propto p(y|X, w)p(w)$$

Key properties:

- Posterior is Gaussian (conjugacy)
- Closed-form solution exists

Interpretation:

- Prior encodes preference for small/structured weights
- Posterior balances data fit and prior belief

Prediction and Connection to Regularization

Predictive distribution:

$$p(y_*|x_*, X, y) = \mathcal{N}(\mu_*, \sigma_*^2)$$

Mean vs variance:

- μ_*: prediction
- σ_*^2: uncertainty due to finite data

Connection to ridge regression:

$$\hat{w}_{\text{MAP}} = \arg\min_w \|y - Xw\|^2 + \lambda\|w\|^2$$

- MAP $\equiv$ Maximum A Posteriori
- Gaussian prior $\leftrightarrow$ ℓ_2 regularization
- Ridge gives only point estimate; Bayesian gives full distribution

Basis Functions and Induced Covariance

Consider nonlinear model:

$$f(x) = w^T \phi(x)$$

Assume:

$$w \sim \mathcal{N}(0, I)$$

Then:

$$\mathbb{E}[f(x)] = 0, \quad \mathrm{Cov}(f(x), f(x')) = \phi(x)^T \phi(x')$$

Key insight:

- Function values are correlated via feature similarity
- Model depends only on inner products

This leads naturally to kernels.

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Kernel Methods as Generalized Similarity

Define:

$$k(x, x') = \phi(x)^T \phi(x')$$

Implications:

- Avoid explicit feature construction
- Work in high or infinite-dimensional spaces

Connection to earlier ideas:

- Kernel smoothing: uses $K(x, x_i)$ heuristically
- Kernel methods: arise from inner product structure

Unifying idea:

Prediction depends on similarity structure encoded by $k(x, x')$

Kernel Ridge Regression

Prediction:

$$\hat{f}(x) = k(x, X)(K + \lambda I)^{-1} y$$

Observations:

- Same algebraic structure as kernel smoothing
- Weights depend on all data points

Interpretation:

- Regularization controls smoothness
- Kernel defines notion of locality

Limitation:

- Still produces deterministic predictions

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From Finite to Infinite Basis Expansions

Recall:

$$f(x) = w^T \phi(x), \quad w \sim \mathcal{N}(0, I)$$

If $\phi(x)$ is finite-dimensional:

- Bayesian linear regression

If $\phi(x)$ is infinite-dimensional:

- Define covariance directly using kernel

Result:

$$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$

Interpretation:

- Prior over functions instead of parameters
- Infinite possible functions fit data
- Need principled way to choose among them

Gaussian Process Definition

$$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$

- $m(x)$: mean function
- $k(x, x')$: kernel (covariance)

Key property:

$$[f(x_1), \ldots, f(x_n)] \sim \mathcal{N}(m, K)$$

$$K_{ij} = k(x_i, x_j)$$

Interpretation:

- Kernel encodes similarity
- Similar inputs $\Rightarrow$ correlated outputs

Kernel Matrix Structure

$$K = \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix}$$

Dimensions:

$$K \in \mathbb{R}^{n \times n}$$

Common kernels:

- RBF: $\exp(-\|x - x'\|^2/2\ell^2)$
- Linear: $x^T x'$
- Polynomial

Joint Gaussian Prior

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{f}_* \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} K(X,X) + \sigma_n^2 I & K(X,X_*) \\ K(X_*,X) & K(X_*,X_*) \end{bmatrix}\right)$$

$$K(X,X) \in \mathbb{R}^{n\times n}, \quad K(X,X_*) \in \mathbb{R}^{n\times n_*}$$

$$p(\mathbf{f}_* \mid X, \mathbf{y}, X_*) = \mathcal{N}(\boldsymbol{\mu}_*, \Sigma_*)$$

Mean:

$$\boldsymbol{\mu}_* = K(X_*,X)\,[K(X,X) + \sigma_n^2 I]^{-1}\mathbf{y}$$

Covariance:

$$\Sigma_* = K(X_*,X_*) - K(X_*,X)\,[K(X,X) + \sigma_n^2 I]^{-1} K(X,X_*)$$

An Example

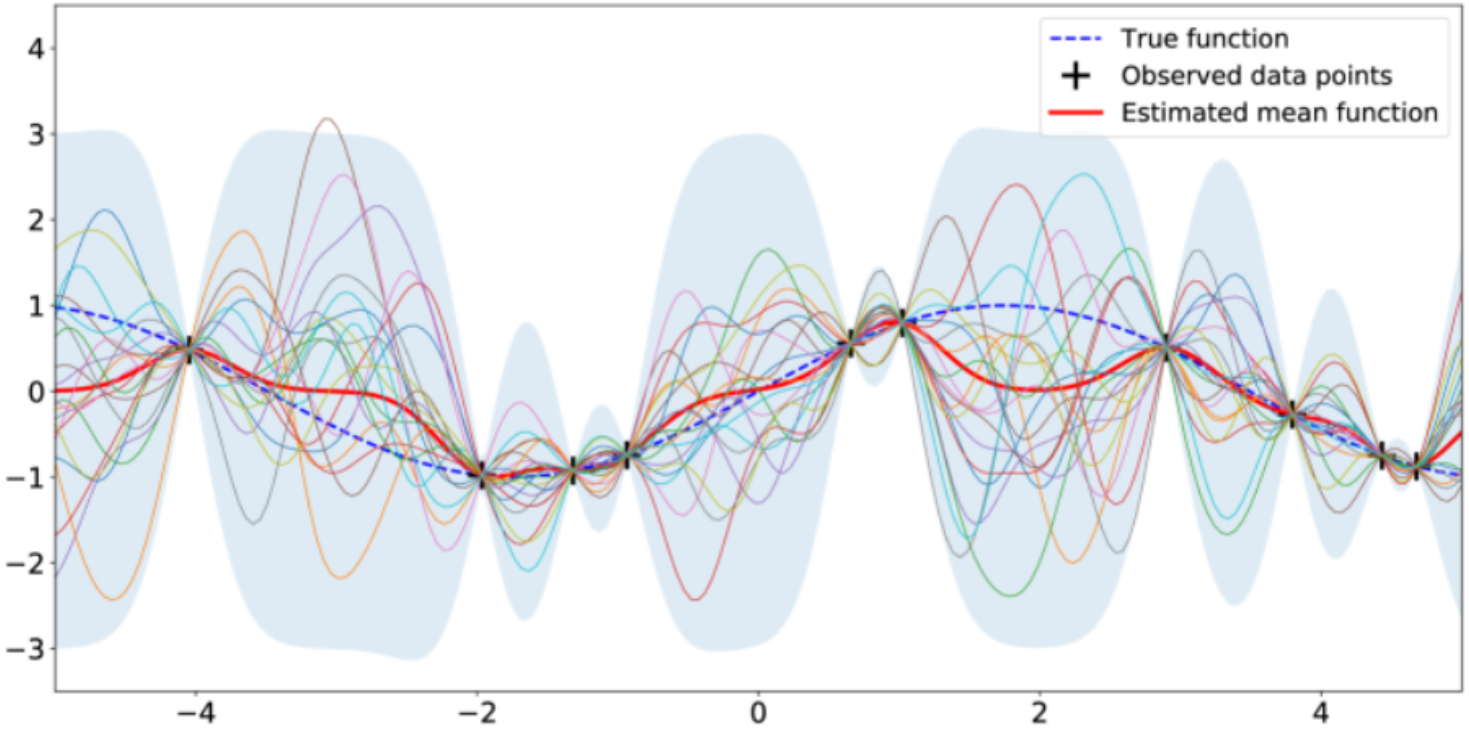


Figure 10: An illustrative example of standard GPR. Black crosses represent observed data points generated by the blue dotted line (true function). Given these data points, infinite possible posterior functions were obtained, with 20 samples plotted in different colors. The mean function, derived from the probability distribution of these functions, is plotted as a red solid line. The blue shaded area around the mean function indicates 3 times prediction variances.

Figure: Enter Caption

Connection to Kernel Ridge Regression

$$\hat{\mathbf{f}}_* = K(X_*, X)\,[K(X, X) + \lambda I]^{-1}\mathbf{y}$$

$$\boldsymbol{\mu}_* = K(X_*, X)\,[K(X, X) + \sigma_n^2 I]^{-1}\mathbf{y}$$

$$\lambda \equiv \sigma_n^2$$

Observation:

$$\hat{\mathbf{f}}_* = \boldsymbol{\mu}_*$$

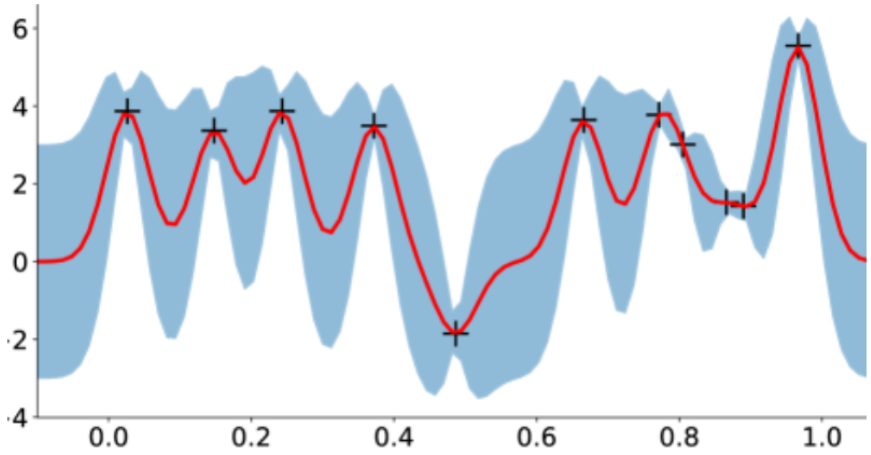
Kernel ridge regression $\equiv$ posterior mean of GP

Structure mirrors kernel ridge regression but adds uncertainty.

Kernel Role

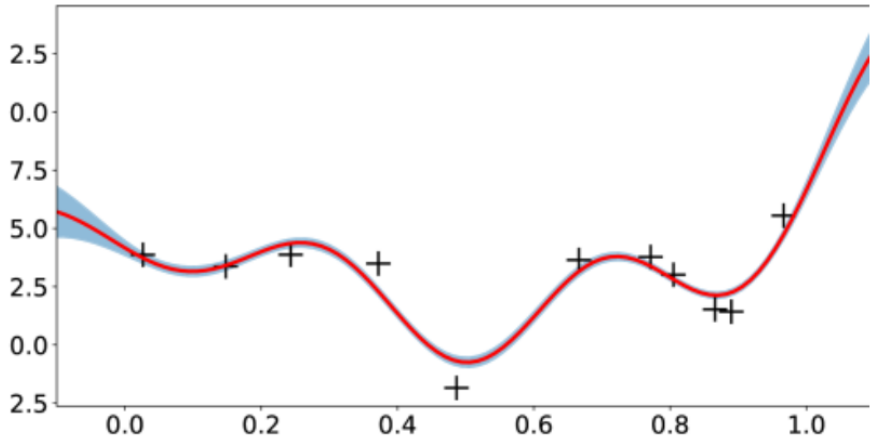
$$k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right)$$

ℓ : length-scale, σ_f^2 : signal variance

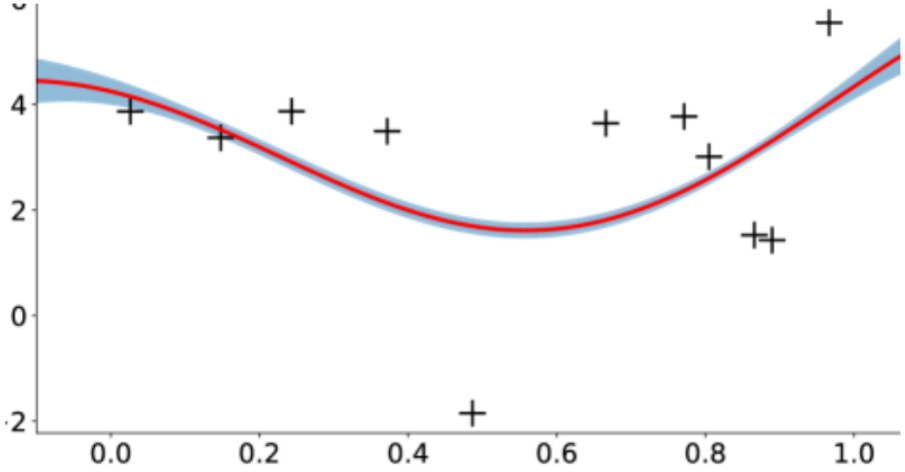


(a) l = small

HP (contd.)



(b) l = medium



(c) l = large

Interpretation: GP as Probabilistic Kernel Smoother

Mean:

- Weighted combination of observed outputs
- Similar to kernel smoothing and kernel ridge regression

Variance:

- Quantifies uncertainty
- Small near observed data, large far away

Key conceptual synthesis:

$$\text{Kernel smoothing} + \text{Bayesian inference} = \text{Gaussian Process}$$

Conceptual Summary

- Kernel smoothing:
  - Local averaging using similarity
  - No probabilistic foundation
- Bayesian regression:
  - Prior on parameters
  - Provides uncertainty
- Gaussian Processes:
  - Prior on functions
  - Kernel defines geometry of function space
  - Unifies locality and uncertainty

Final insight:

GPs are the natural probabilistic extension of kernel methods.

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From Kernels to Learned Similarity (Attention)

Kernel methods use a fixed similarity:

$$k(x, x') = \phi(x)^T \phi(x')$$

Limitation: Similarity is predefined; Does not adapt to data representation
Modern approach (Attention):

$$\text{Attn}(x, x_i) = \frac{\exp(q(x)^T k(x_i))}{\sum_j \exp(q(x)^T k(x_j))}$$

- Similarity is learned via neural networks
- Depends on input-dependent representations $q(x), k(x_i)$

Connection:

- Kernel smoothing $\Rightarrow$ fixed weights from $k(x, x_i)$
- Attention $\Rightarrow$ learned, adaptive weighting

Deep Kernel Learning

Standard kernel:

$$k(x, x') = k_{\text{RBF}}(x, x')$$

Generalization:

$$k(x, x') = k_\theta(\phi_\theta(x), \phi_\theta(x'))$$

- $\phi_\theta(x)$: neural network representation
- Kernel operates in learned feature space

Advantages:

- Captures complex structure in data
- Retains GP uncertainty quantification

Interpretation:

- Combines representation learning + probabilistic modeling

Scalability of Gaussian Processes

Exact GP inference requires:

$$(K + \sigma^2 I)^{-1}, \quad \mathcal{O}(n^3)$$

Limitation:

- Not scalable to large datasets

Sparse approximation: Introduce inducing points $Z = \{z_1, \ldots, z_m\}$, $m \ll n$

$$K(X, X) \approx K(X, Z)K(Z, Z)^{-1}K(Z, X)$$

Effect:

- Reduces computation to $\mathcal{O}(nm^2)$
- Enables large-scale GP models

Deep Gaussian Processes

Single GP:

$$f(x) \sim \mathcal{GP}(0, k(x, x'))$$

Composition:

$$f_L(x) = f_L(f_{L-1}(\cdots f_1(x)))$$

- Each layer is a GP
- Outputs of one GP become inputs to next

Motivation:

- Standard GP is limited to one layer of transformation
- Deep structure allows hierarchical feature learning

Connection:

- Analogous to deep neural networks

Uncertainty in Modern Models

Gaussian Processes:

- Provide exact posterior uncertainty
- Variance reflects data density and noise

Neural networks:

- Typically produce point estimates
- Often overconfident outside training data

Approximate Bayesian methods:

- Monte Carlo Dropout
- Deep Ensembles
- Bayesian Neural Networks

Key point:

- GPs remain a gold standard for calibrated uncertainty